

SEMESTER-IV

URA402: ROBOTICS SENSORS AND ACTUATORS				
	L	T	P	Cr
	2	0	2	3.0
Course Objective: To provide students with an understanding of the design, analysis, and application of sensors and actuators for dynamical systems. Students will learn about different types of sensors and actuators, their operating principles, and the mechanisms used to convert physical signals into electrical signals and vice versa. Through laboratory experiments and projects, students will learn to select, design, and analyze sensors and actuators for different applications. Additionally, students will learn to integrate sensors and actuators into mechanical systems.				
Syllabus Introduction: Classification of sensors and actuators, performance characteristics: least square polynomials and data fitting, impedance and impedance matching, static and dynamic characteristics, frequency response, calibration, excitation, deadband, reliability. Interfacing methods and circuits: Case studies: Interfacing of sensors and actuators with microprocessor or microcontroller. Signal conditioning circuits: Op-Amp based signal conditioning circuits, Integrator, differentiator, Difference amplifier, Instrumentation amplifier. DC Bridges, use of Wheatstone bridge for resistive transducers. Active Filters in instrumentation circuits: High pass low pass, Band pass, Design of Butterworth and Chebyshev filters. Data Acquisition Techniques: Sample and hold circuits, Sampling and quantization, Interference, Analog to Digital convertors (ADC): ramp type and SAR, and Digital to Analog Convertors (DAC) R-2R based DAC, Multiplexer and de-multiplexer. Sensors for manipulator systems: shaft encoders, potentiometers, strain gauge, piezoelectric accelerometers, LVDTs. Sensors for autonomous mobile robots: Navigation Sensors: odometry, accelerometer, gyrometer, magnetometer, barometer, Inertial Measurement Unit, GPS, Proximity Sensors: ultrasonic sensor, LiDAR (2D & 3D). Actuators: Electric Linear, Electric Rotary, Fluid Power Linear, Fluid Power Rotary, Linear Chain Actuators, Manual Linear, Manual Rotary, soft actuators				
Laboratory Work Measurement of Linear Displacement, Angular displacement, Temperature, Thermistor and RTD for temperature measurement. Measurement of displacement using LVDT and Capacitive transducer, Position and velocity measurement using encoders. RTDs, thermocouples, capacitive accelerometers, gyrometer, and LiDAR. human-motion capture, force sensor.				
Course Learning Objectives (CLO) The students will be able to:				

Approved in 113th meeting of the Senate held on September 7, 2024, respectively. Further, revised in 1st meeting of the Academic Council held on September 11, 2025.

1. select suitable sensors for measuring physical parameters.
2. elucidate the interfacing of sensors for robotics and automation.
3. elucidate the working of ADCs and DACs.
4. validate the working of sensors and actuators.

Text Books

1. Sensors, Actuators, and Their Interfaces, Nathan, I., IET 2nd edition, 2020
2. Sensors and Transducers, Patronabis, D., Prentice Hall India, 2011.
3. Measurement Systems: Applications and Design, Doebelin, E.O. and Manic, D.N., 7th edition, McGraw-Hill, 2019.

Reference Books

1. Process Dynamics and Control, Seborg, D.E. and Edgar, T., John Wiley and Sons, 1989
2. Transducers and Instrumentation, Murthy, D.V.S., Prentice Hall of India, 2003.
3. Instrumentation, Measurement and Analysis, Nakra, B.C. and Chaudhry, K.K., Tata McGraw Hill, 2003.
4. Micromechanics and MEMS, Trimmer, W., IEEE Press, 1990

Evaluation Scheme:

Sr. No.	Evaluation Elements	Weightage (%)
1	MST	25
2	EST	40
3	Sessional (May include Assignments/Projects/Tutorials/Quizzes/Lab Evaluations)	35

Approved in 113th meeting of the Senate held on September 7, 2024, respectively. Further, revised in 1st meeting of the Academic Council held on September 11, 2025.